TECHNICAL ARCHITECTURE & EXECUTIVE PROJECT DOSSIER

Automated Tabular Data Cleansing & Executive Report Generator

Author: Anurag Kumar Poddar (Data Analyst | Python | SQL | Power BI | Data Visualization)

Core Class: CorporateReportBuilder

.CSV / .XLSX / .XLS	≥ 60% Heuristic	6 Aggregations	200 DPI PNG
Multi-Format Ingestion	Smart Numeric Detection	Sum, Mean, Median, Min, Max, Count	Executive Chart Export

1. Executive Overview & Problem Statement

The **Automated Tabular Data Cleansing & Executive Report Generator** (`CorporateReportBuilder`) is a Python-based desktop application engineered to streamline ad-hoc data analysis and executive reporting. In corporate operational environments, raw datasets frequently suffer from critical formatting inconsistencies—such as numeric values stored as text strings with comma separators, unstandardized categorical text labels, and corrupted or missing values. This application automates the end-to-end ingestion, sanitization, aggregation, and visualization of raw tabular files (`.csv`, `.xlsx`, `.xls`), empowering business users and analysts to generate clean summary tables and publication-ready charts through an intuitive graphical interface without writing a single line of code.

2. Core Functional Pillars & Cleansing Pipeline

- **Intelligent Data Ingestion & Heuristic Type Detection:** Seamlessly loads CSV and Microsoft Excel workbooks while extracting instant metadata profiles (row/column dimensions and column header inventories). Goes beyond standard type checking by scanning string (`object`) columns and heuristically identifying numeric-like columns where at least 60% of records (or a minimum threshold of 3 rows) can be parsed as valid numbers after stripping formatting.
- **Automated Data Cleansing Pipeline:** Standardizes categorical grouping columns by stripping leading and trailing whitespaces and converting text to `Title Case` to eliminate duplicate grouping keys. Executes safe numeric coercion by stripping comma separators (converting `"12,000"` to `12000.0`), coercing corrupted entries to `NaN`, and dropping null records prior to calculation.
- **Dynamic Aggregation & Live Treeview Preview:** Provides interactive dropdown controls for categorical GroupBy dimensions, numeric target metrics, and six statistical functions (**Sum, Mean/Average, Median, Max, Min, Count**). Results are automatically sorted in descending order and displayed in a scrollable `Treeview` table formatted to two-decimal precision with thousand separators (`1,234.56`).
- **Executive Chart Builder & One-Click Export:** Embeds an interactive Matplotlib canvas (`FigureCanvasTkAgg`) supporting Horizontal Bar, Vertical Column, Line (with markers), and Pie charts (with percentage labels). Automatically slices Bar/Column/Line charts to the **Top 10** categories and Pie charts to the **Top 6** categories for visual clarity, exporting reports to `.xlsx`/`.csv` and visuals to `200 DPI .png` files.

3. End-to-End Processing Pipeline & Tech Stack

FIGURE 1: AUTOMATED DATA INGESTION, SANITIZATION & REPORTING PIPELINE (PYTHON 3.X | TKINTER | PANDAS | MATPLOTLIB)



4. Software Architecture: CorporateReportBuilder Class

Method Signature	Subsystem Layer	Technical Responsibility
`_build_ui()`	GUI Presentation (Tkinter)	Initializes window geometry (`1100x700`), file controls, `Combobox` selectors, scrollable `Treeview`, and Matplotlib container.
`_safe_numeric_convert()`	Sanitization Utility	Strips comma characters from string series and safely coerces values to floats via `pd.to_numeric(..., errors="coerce")`.
`browse_file()` / `read_file()`	I/O & Heuristic Profiler	Loads CSV/Excel files via `pandas`/`openpyxl`, outputs shape/columns, and classifies text vs. numeric-like columns.
`preview_report()` / `_render_table()`	Cleansing & GroupBy Engine	Applies `.str.strip().str.title()`, drops nulls, aggregates metrics, sorts descending, and formats floats (`1,234.56`).
`preview_chart()`	Visual Analytics Engine	Slices top 10 (Bar/Column/Line) or top 6 (Pie) categories and renders figures onto `FigureCanvasTkAgg`.
`export_report()` / `export_chart()`	Persistence & Export	Writes cleansed tables to `report_output.xlsx/.csv` and saves 200 DPI charts to `chart_output.png` in the source directory.

5. Operational Workflow & Future Roadmap

1. **Select & Read File:** Click *Browse* to choose a local `.csv`, `.xlsx`, or `.xls` dataset, then click *Read* to inspect row/column dimensions and auto-populate column dropdowns.
2. **Configure & Preview Report:** Select the categorical *Group By* column, statistical *Aggregation*, and numeric *Value* metric. Click *Preview Report* to cleanse records and populate the sorted table, then click *Export Report*.
3. **Generate & Export Visuals:** Select a chart style (`Bar`, `Column`, `Pie`, or `Line`), click *Preview Chart* to inspect on the embedded canvas, and click *Export Chart (PNG)* to save at 200 DPI.

> **Planned Future Enhancements & Scalability Roadmap**
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> - **Multi-Level Grouping:** Support hierarchical aggregation across multiple categorical dimensions